

Presentation Title

Presentation Subtitle and Description

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1. Introduction
 - 1.1 Topic #1
 - 1.2 Topic #2
2. Methodology
 - 2.1 Topic #1
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3. Conclusion and Outlook
 - 3.1 Topic #1
 - 3.2 Topic #2

Core Concepts:

- ▶ Quantum systems interact with their environment, leading to decoherence.
- ▶ Markovian noise assumes the environment has no memory (information flows strictly from system to environment).
- ▶ **Non-Markovian noise** involves information backflow, where the environment retains memory of the system's past states.

Implications for Error Correction:

1. Standard Quantum Error Correction (QEC) assumes independent, memoryless errors.
2. Memory effects can cause correlated errors over time, degrading threshold theorems.

System Overview:

- ▶ The setup involves a superconducting qubit coupled to a readout resonator.
- ▶ We induce memory effects by strongly coupling the qubit to a secondary lossy cavity.
- ▶ Measurements are taken using standard Ramsey interferometry.

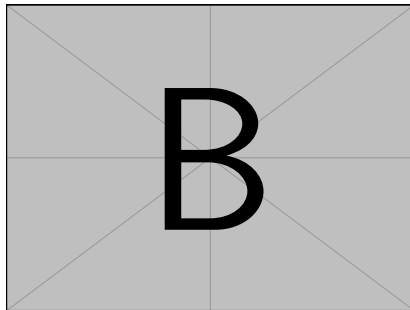


Fig 1: Schematic of the qubit-cavity architecture.

Fidelity Decay Over Time

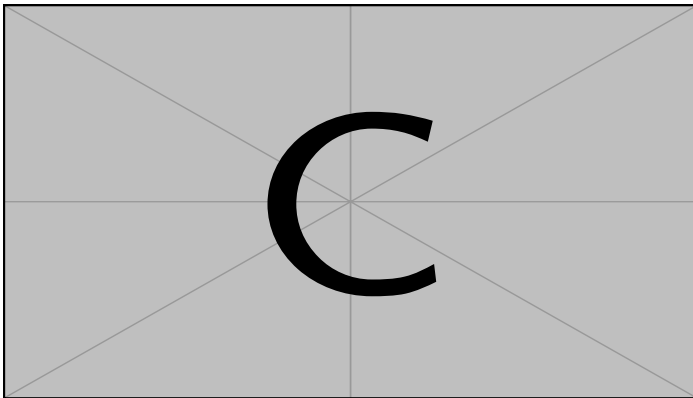


Fig 2: The revivals in fidelity clearly indicate information backflow characteristic of non-Markovian dynamics.

To describe the evolution of the open quantum system, we use the time-local master equation rather than the standard Lindblad form.

Time-Convolutionless (TCL) Master Equation:

$$\frac{d}{dt}\rho(t) = \mathcal{K}(t)\rho(t)$$

$$\mathcal{K}(t)\rho(t) = -i[H(t), \rho(t)] + \sum_i \gamma_i(t) \left(L_i \rho(t) L_i^\dagger - \frac{1}{2} \{L_i^\dagger L_i, \rho(t)\} \right)$$

- ▶ Here, $\gamma_i(t)$ represents the time-dependent decay rates.
- ▶ A signature of non-Markovianity is when $\gamma_i(t) < 0$ for some interval of time, indicating information flowing back into the system.

Trace Distance:

A common measure for distinguishability between two quantum states ρ_1 and ρ_2 is the trace distance:

$$\begin{aligned} D(\rho_1, \rho_2) &= \frac{1}{2} \|\rho_1 - \rho_2\|_1 \\ &= \frac{1}{2} \text{Tr} \left(\sqrt{(\rho_1 - \rho_2)^\dagger (\rho_1 - \rho_2)} \right) \end{aligned}$$

When $\frac{d}{dt} D > 0$, the states become more distinguishable, which is our signature for non-Markovianity.

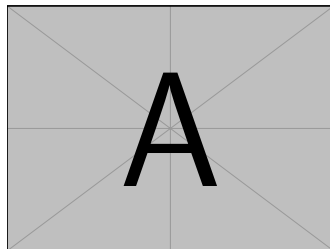


Fig 3: Trace distance oscillation over time.

- [1] Breuer, H. P., Laine, E. M., & Piilo, J. (2009). *Measure for the degree of non-Markovian behavior of quantum processes in open systems*. Physical Review Letters, 103(21), 210401.
- [2] Reference Two (e.g., specific QEC paper).

Thank You!

Questions?

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